

An integrated polarization bound without exponential coefficient loss

Independent asymptotic companion calculation

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Use the notation of `forced_success_mixture_comparison.tex`: $\ell = \lfloor m/4 \rfloor$, $N = m - \ell$, $\alpha = \ell/m$, $\delta_j = 1 - p_j$, $s = \sum \delta_j$, $x_j = \ell \delta_j$, $A_B = \prod_{j \in B} p_j$, and $x_R = N^{-1} \sum_{j \in R} x_j$ for $R = B^c$. We use the established estimates

$$\max_j \delta_j \leq 256(s+1)/m, \quad \mathbb{E}_B A_B \leq e^{-\alpha s}, \quad \mathbb{E}_w e^{-cs} \leq C_c/m \quad (c > 0 \text{ fixed}). \quad (1)$$

Fix $0 < \theta \leq 1/16$. The polynomials R_d and P_d below are the scaled Laguerre taper and normalized square from that companion note. Their degrees are at most $2d - 1$, and we require $2d \leq N$.

Lemma 1 (Centered elementary symmetric coefficients). *If $y_1, \dots, y_N$ are real, $\sum y_j = 0$, and $|y_j| \leq D$, then*

$$\frac{|e_k(y)|}{\binom{N}{k}} \leq \left(\sqrt{e} D \sqrt{k/N} \right)^k \quad (1 \leq k \leq N). \quad (2)$$

Proof. The elementary inequality $|1 + u| \leq \exp(\operatorname{Re} u + |u|^2/2)$ gives, for any complex z ,

$$\left| \prod_j (1 + y_j z) \right| \leq \exp \left(|z|^2 \sum_j y_j^2 / 2 \right).$$

Cauchy's coefficient estimate at radius $\sqrt{k / \sum y_j^2}$ yields $|e_k(y)| \leq (e \sum y_j^2 / k)^{k/2}$; the case of zero sum of squares is immediate. Now use $\sum y_j^2 \leq ND^2$ and $\binom{N}{k} \geq (N/k)^k$. $\square$

For a polynomial P of degree at most N , write $\operatorname{Pol}_R(P)$ for its symmetric multiaffine polarization on the N variables indexed by R . We have the exact

expansion

$$\text{Pol}_R(P)(x_j : j \in R) - P(x_R) = \sum_{k=2}^{\deg P} \frac{P^{(k)}(x_R)}{k!} \frac{e_k(x_j - x_R : j \in R)}{\binom{N}{k}}. \quad (3)$$

The sum starts at two because the centered variables sum to zero.

Lemma 2 (Integrated error). *For $d \geq 16$ with $d \leq c\sqrt{m}$ and $2d \leq N$, where $c > 0$ is a sufficiently small absolute constant depending only on fixed θ ,*

$$\begin{aligned} (\ell + 1)\mathbb{E}_w\mathbb{E}_B[A_B|\text{Pol}_R(R_d) - R_d(x_R)] &\leq Cd^4/m, \\ (\ell + 1)\mathbb{E}_w\mathbb{E}_B[A_B|\text{Pol}_R(P_d) - P_d(x_R)] &\leq Cd^2/m. \end{aligned}$$

Proof. The standard Laguerre bound $|L_j^{(k)}(u)| \leq \binom{j}{k}e^{u/2}$ for $u \geq 0$ implies, by Leibniz and Vandermonde's identity,

$$\frac{|(q_d(\theta u)^2)^{(k)}|}{k!} \leq d^2 e^{\theta u} \frac{(4\theta d)^k}{(k!)^2}. \quad (4)$$

Indeed the Leibniz sum for $L_i L_j$ is at most $2^k \sum_a \binom{i}{a} \binom{j}{k-a} = 2^k \binom{i+j}{k}$. For $R_d(u) = (2/(\theta d) - u)q_d(\theta u)^2$, differentiating its linear factor in addition shows

$$\frac{|R_d^{(k)}(u)|}{k!} \leq C_\theta d^2 (1+u) e^{\theta u} \frac{(C_\theta d)^k (1+k^2)}{(k!)^2}. \quad (5)$$

For $P_d = h^{-2}(\sum_{j < h} L_j(\theta u))^2$, $h = \lfloor d/8 \rfloor$, the corresponding bound is

$$\frac{|P_d^{(k)}(u)|}{k!} \leq C_\theta e^{\theta u} \frac{(C_\theta d)^k}{(k!)^2}. \quad (6)$$

At every row and for every B , (1) gives $|x_j - x_R| \leq C(s+1)$, while $x_R \leq s/3$. Insert (2), (5) into (3), multiply by A_B , and average over B . Since $\mathbb{E}_B A_B \leq e^{-\alpha s}$ and $\alpha \geq 1/5$, the result is bounded by

$$Cd^2 e^{-c_0 s} \sum_{k \geq 2} \left(\frac{Cd}{\sqrt{m}} \right)^k \frac{k^{k/2} (1+k^2)}{(k!)^2} (1+s)^{k+1}, \quad (7)$$

where $c_0 > 0$ is fixed. Extending the finite sum to infinity is harmless.

For each fixed nonnegative integer a , (1) also gives the explicit factorial estimate

$$\mathbb{E}_w[(1+s)^{k+a} e^{-c_0 s}] \leq \frac{C^{k+a+1} (k+a)!}{m}. \quad (8)$$

To see this, bound $(1+s)^{k+a}e^{-c_0s/2}$ by $C^{k+a}(k+a)!$ and apply the Laplace estimate to the remaining exponential. After multiplying (7) by $\ell+1 = O(m)$ and using (8), the total error is at most

$$Cd^2 \sum_{k \geq 2} \left(\frac{Cd}{\sqrt{m}} \right)^k \frac{k^{k/2}(1+k^2)(k+1)!}{(k!)^2}.$$

Since $k! \geq (k/e)^k$, the coefficient apart from the power is at most $C^k(1+k)^3/\sqrt{k!}$. For $Cd/\sqrt{m} \leq 1$ the series starting at $k=2$ is $O(d^2/m)$. This proves the first assertion. The same argument with (6), and $a=0$, removes the prefactor d^2 and proves the second. $\square$

The two factorials in (4) are essential to this estimate. Bounding polynomial coefficients separately would instead introduce an exponential factor in d and recover only the earlier logarithmic scale. This lemma supplies the polarization part of a prospective power lower-bound proof; the exact beta identities, scalar test positivity, and mixture comparison are separate inputs.

Variance-sensitive upgrade

The stronger off-diagonal Gaussian bound proved in the main power lower-bound note is

$$z^{h/2} |(L_j(\theta z))^{(h)}| \leq (\theta j)^{h/2} e^{\theta z/2} \quad (z > 0). \quad (9)$$

Equivalently it follows from the modulus bound on the unitary Gaussian translation matrix coefficient $e^{-u/2} u^{h/2} \sqrt{(j-h)!/j!} L_{j-h}^{(h)}(u)$. The following calculation is conditional only on that explicitly stated classical inequality, in addition to (1).

Lemma 3 (Variance-sensitive integrated polarization). *Let $Q_a(z) = \sum_{j=0}^d c_{aj} L_j(\theta z)$, $B_a = \sum_j |c_{aj}|$, $a=1,2$, and put $P = Q_1 Q_2$. For $2d \leq N$ and $d \leq cm$, with a sufficiently small fixed $c > 0$,*

$$(\ell+1) \mathbb{E}_w \mathbb{E}_B [A_B | \text{Pol}_R(P) - P(x_R)] \leq C B_1 B_2 \frac{d}{m}. \quad (10)$$

Proof. The proof of (2), before replacing the sum of squares by ND^2 , gives the stronger bound

$$\frac{|e_k(y)|}{\binom{N}{k}} \leq \left(\frac{\sqrt{e} \sigma \sqrt{k}}{N} \right)^k, \quad \sigma^2 = \sum_j y_j^2. \quad (11)$$

By (9) and the product rule, for $z > 0$,

$$\frac{|P^{(k)}(z)|}{k!} \leq B_1 B_2 e^{\theta z} \frac{(2\sqrt{\theta d/z})^k}{k!}. \quad (12)$$

At a fixed row and fixed B , write $v = x_R$ and $y_j = x_j - v$. If $v = 0$, all x_j are zero, so the polarization error is zero. Otherwise

$$\sigma^2 \leq \sum_{j \in R} x_j^2 \leq (\max_j x_j) \sum_{j \in R} x_j \leq CN(s+1)v. \quad (13)$$

Insert (11)–(13) into the exact expansion (3). The powers of v cancel. After averaging with A_B and using (1), the result is bounded by

$$CB_1 B_2 e^{-c_0 s} \sum_{k=2}^{2d} \left(C\sqrt{d/m} \right)^k \frac{k^{k/2}}{k!} (s+1)^{k/2}.$$

The factorial weighted-moment estimate (8), with $\lceil k/2 \rceil$ in place of $k+a$, shows that the normalized integrated error is at most

$$CB_1 B_2 \sum_{k=2}^{2d} \left(C\sqrt{d/m} \right)^k \frac{k^{k/2} \lceil k/2 \rceil!}{k!}.$$

For even and odd k separately, the elementary factorial bounds $(k/e)^k \leq k! \leq k^k$ give

$$\frac{k^{k/2} \lceil k/2 \rceil!}{k!} \leq C^k (k+1).$$

Thus the series is geometric with a polynomial factor, starts at $k=2$, and is $O(B_1 B_2 d/m)$ whenever $d \leq cm$ with c sufficiently small. This proves (10). $\square$

For the endpoint taper, the recurrence identity $yq_d(y) = d^{-1} \sum_{j < d} L_j(y) - L_d(y)$ factors $R_d = q_d(\theta z)[(2/(\theta d) - z)q_d(\theta z)]$ into two Laguerre expansions with coefficient sums $O(d)$ and $O(1)$. Hence (10) bounds its polarization error by $O(d^2/m)$. This is compatible with endpoint degree d of order $\sqrt{m}$; the weighted-mixture comparison must satisfy the corresponding bound as a separate input.